

Other Endocrine Causes of Hypertension

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Endocrine causes of hypertension often are undiagnosed. Some studies suggest endocrine causes make up over 20% of all secondary hypertension cases, of which primary aldosteronism is most common.¹

Endocrine hypertension frequently occurs in the absence of readily observed signs and symptoms, but certain features should trigger consideration (Fig. 40.1). Chapter 39 discusses primary aldosteronism; this chapter describes the other causes of endocrine-related hypertension.

CUSHING SYNDROME

Definition

Cushing syndrome is defined by sustained glucocorticoid excess that does not follow a circadian rhythm. Cushing syndrome can be caused by overproduction of adrenocorticotrophic hormone (ACTH) by a pituitary adenoma (i.e., Cushing disease); by cortisol overproduction from an adrenal adenoma, adrenal nodular hyperplasia, or adrenal cortical carcinoma; or rarely as secondary to ectopic ACTH (corticotrophin) secretion from neuroendocrine tumors.²⁻⁴ Approximately 0.5% of patients with bronchogenic carcinoma (more common in men than women) develop ectopic ACTH syndrome. Even more rare, ectopic corticotropin-releasing hormone (CRH) from tumors can cause hypercortisolism. Another rare cause of Cushing syndrome is Carney complex, an autosomal dominant disorder featuring cardiac myxomas and pigmented skin/mucosal lesions and Cushing syndrome. The most common cause of Cushing syndrome is exogenous corticosteroid administration, including from steroids present in joint injections, creams, inhalers, nasal sprays, and oral steroids. The incidence of endogenous Cushing syndrome is 5 to 10 cases per 1 million population per year. Cushing disease and cortisol-secreting adrenal tumors are four times more common in women than in men.

Etiology, Pathogenesis, and Epidemiology

Hypertension is present in 80% of patients with Cushing syndrome (less often when caused by exogenous synthetic corticosteroid administration) and results from an increase in both cardiac output and total peripheral resistance.⁵

In some patients, Cushing syndrome may be caused by concurrent overproduction of mineralocorticoids such as aldosterone, 11-deoxycorticosterone, and corticosterone. Although cortisol can bind to the mineralocorticoid receptor (MR) in the kidney, it usually does not because 11 β -hydroxysteroid dehydrogenase type 2 (11 β -HSD2) inactivates cortisol to cortisone, thereby preventing its binding to the MR. However, in patients with extremely high cortisol levels (e.g., severe Cushing syndrome as in ectopic ACTH syndrome), 11 β -HSD is overwhelmed, and excess cortisol can bind the MR and lead to a pseudohyperaldosterone state.

Cardiac and vascular MR activation may foster cardiovascular inflammatory, hypertrophic, and fibrotic changes. Inhibition of

the vasodilator nitric oxide (NO) by cortisol may contribute to the hypertension, along with enhanced pressor responsiveness to catecholamines and angiotensin II (Ang II), heightened cardiac inotropic sensitivity to β -adrenergic stimulation, and increased plasma volume.⁵ The sympathetic nervous system and renin-angiotensin system (RAS) may be suppressed, even though circulating levels of renin substrate are increased. Adipokines, including leptin and resistin, plus release of proinflammatory cytokines (tumor necrosis factor- α , interleukin-6), also may contribute to the increased cardiovascular (CV) risk observed in Cushing syndrome.^{2-4,6-8}

Successful treatment of Cushing disease or removal of an underlying adrenal adenoma usually results in blood pressure (BP) reduction and partial return of the previously impaired nocturnal fall in arterial pressure, although hypertension persists in some patients.⁸

Clinical Manifestations

Clinical features in Cushing disease result from elevated circulating levels of hormones, including ACTH (increased pigmentation) and cortisol (central adiposity, insulin resistance or diabetes, muscle wasting/weakness, plethoric facies, purple striae [Fig. 40.2], easy bruising, osteoporosis, psychological problems). In some patients, androgen effects are observed (hirsutism, acne, virilization) and may be striking in those with adrenal adenoma or adrenal cortical carcinoma. Ectopic ACTH syndrome caused by small cell lung carcinoma or other tumors (e.g., bronchial or thymic neuroendocrine tumors) manifests typically as a wasting disease, often with hyperpigmentation (from the very high ACTH levels) and hypokalemia (from the MR binding in very high cortisol states). Hypertension is often associated with left ventricular hypertrophy, which can be disproportionate to the blood pressure, and frank cardiac failure is occasionally the presenting feature.^{2,5}

Differential Diagnosis

Pseudo-Cushing syndrome can occur with a sustained high intake of alcohol by inducing augmented cortisol secretion and reduced cortisol metabolism caused by hepatic damage. Routine diagnostic tests are unable to distinguish alcoholic pseudo-Cushing syndrome from true Cushing syndrome, and reassessment after alcohol withdrawal may be required.⁹ Depression is associated with an increased cortisol level. Careful psychological and physical evaluation will usually differentiate these patients from those with Cushing syndrome.

Diagnosis

Initial evaluation should test for one or more of the following: (1) an elevated 24-hour urinary free cortisol excretion (at least two collections), (2) increased late night salivary cortisol (at least three samples), (3) absence of suppression of 7 to 8 AM serum cortisol after a 1 mg dose of dexamethasone taken at midnight the night before the morning blood draw (1 mg dexamethasone suppression test), and (4) failure

Testing for Endocrine Hypertension

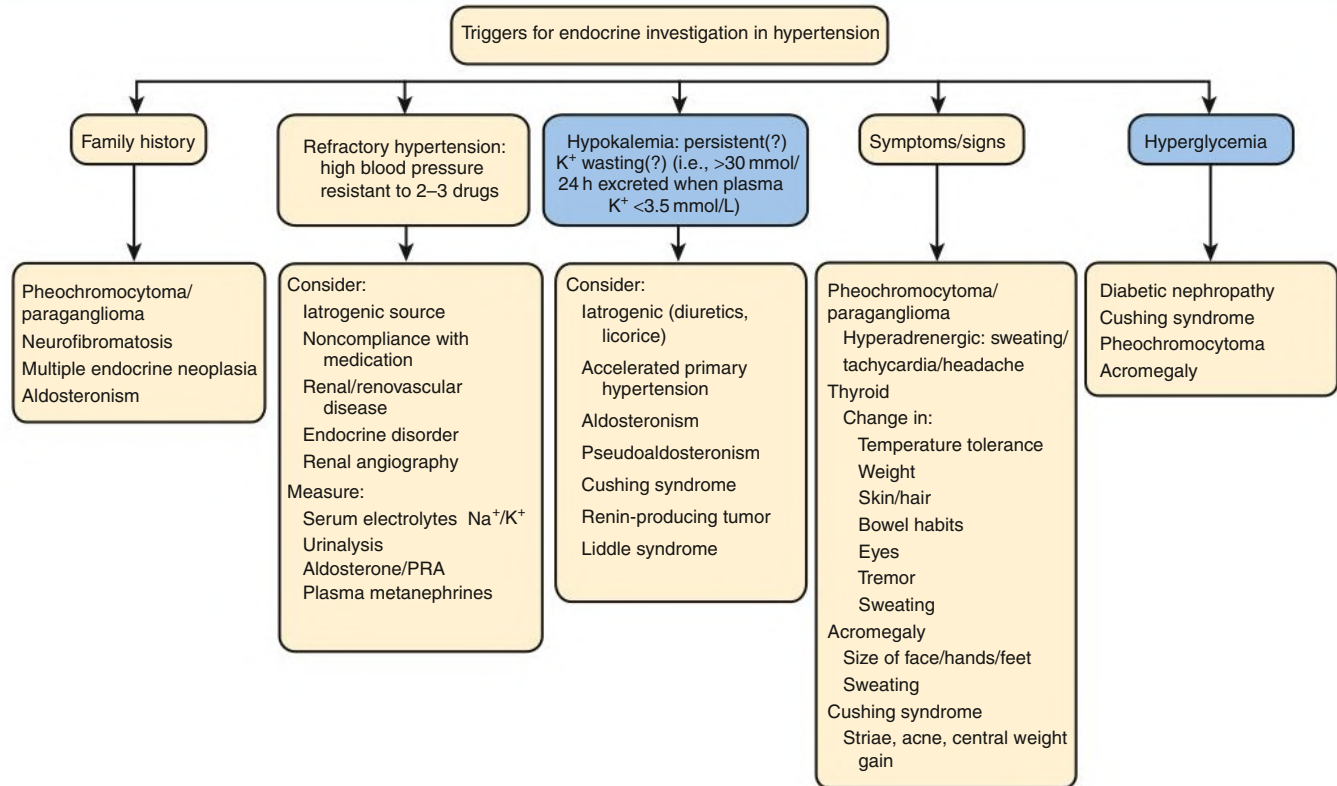


Fig. 40.1 Testing for Endocrine Hypertension. Clinical observations suggesting investigation into endocrine causes of hypertension in hypertensive patients.



Fig. 40.2 Striae and central obesity in a patient with Cushing syndrome.

of suppression of serum cortisol on low-dose dexamethasone (0.5 mg four times daily) for 48 hours. Random blood cortisol measurements are not reliable.

A positive screening test result should trigger involvement of an endocrinologist. Once the screening tests suggest hypercortisolism, early morning cortisol, ACTH, and dehydroepiandrosterone sulfate (DHEAS) can be measured; high levels of ACTH suggest ACTH-dependent causes of hypercortisolism (pituitary adenoma or ectopic ACTH), whereas low ACTH suggests ACTH-independent causes (adrenal adenomas or adrenal cortical carcinoma). Magnetic resonance imaging (MRI) of the pituitary is recommended for ACTH-dependent causes, and computed tomography (CT) of the adrenal glands is recommended for ACTH-independent causes. ACTH-producing pituitary adenomas, corticotroph adenomas, are often small and may not

be well visualized on MRI. For ACTH-dependent cases, further tests may include a high-dose dexamethasone suppression test (8 mg dexamethasone suppression test) that partially suppresses ACTH (at least 50% reduction) in patients with pituitary tumors but not with ectopic ACTH. The response of plasma ACTH to a dose of corticotrophin-releasing hormone ($\geq 20\%$ rise in response to CRH points to a pituitary source) may help distinguish pituitary from adrenal or ectopic tumors. The gold standard to differentiate Cushing disease from ectopic ACTH production is bilateral inferior petrosal sinus sampling (IPSS) for ACTH measurements performed by experienced interventional radiologists. IPSS will show higher ACTH from the petrosal sinus sampling compared with the peripheral blood levels if the source is from the pituitary; no gradient will be seen between the petrosal sinus and the peripheral blood if the source is ectopic. If an ectopic source is suspected, imaging of the thorax/abdomen/pelvis is indicated to detect the most common sources of ectopic ACTH (e.g., lung cancer, pancreatic neuroendocrine tumors, pheochromocytomas/paragangliomas, medullary thyroid cancers). If cross-sectional imaging is unhelpful, ^{68}Ga -DOTATATE positron emission tomography (PET)/CT may be used to highlight tissues with somatostatin receptors and identify the location of ectopic ACTH production.¹⁰

Treatment and Prognosis

Untreated patients with Cushing syndrome have 50% 5-year mortality because of CV risk from hypertension, along with glucose intolerance, insulin resistance, hyperlipidemia, and obesity.^{4,5}

Surgical removal of a pituitary microadenoma in Cushing disease results in 80% to 90% cure rates, whereas 50% cure rates occur for

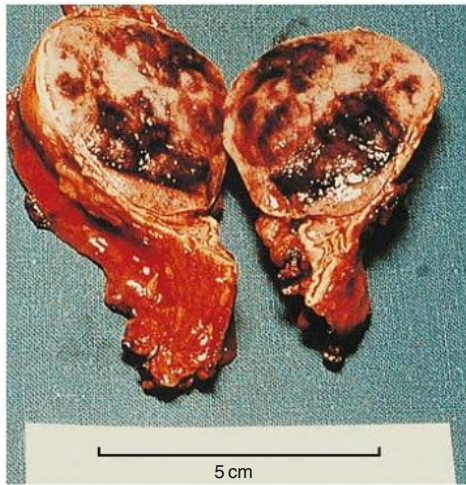


Fig. 40.3 Large adrenal pheochromocytoma with areas of hemorrhagic necrosis.

pituitary macroadenomas. Cushing syndrome due to adrenal adenoma is almost always cured by unilateral adrenalectomy. However, prognosis is poor when Cushing syndrome is from adrenocortical carcinoma or ectopic causes such as small cell lung cancer. If the ectopic ACTH-producing tumor can be located, and complete removal occurs, there is a substantial reduction of mortality. After cure of Cushing syndrome, approximately 30% of patients have persistent hypertension.¹¹

No evidence supports the use of any antihypertensive class over another. Potassium-losing diuretics can exacerbate both hypokalemia and glucose intolerance, whereas potassium-sparing diuretics, usually in combination with other antihypertensive agents, may correct hypokalemia and reduce edema while lowering BP.

PHEOCHROMOCYTOMA/PARAGANGLIOMA

Definition

Pheochromocytoma and paraganglioma (PPGLs) can mimic a wide spectrum of other disorders, and the diagnosis is frequently delayed or missed.¹² PPGLs are tumors of the autonomic nervous system that arise from chromaffin tissue in the adrenal medulla (Fig. 40.3) and from sympathetic extra-adrenal ganglia, which secretes catecholamines and metanephrines, or the parasympathetic ganglia, which are nonsecretory, especially those in the head and neck (Fig. 40.4).

PPGLs occur in 2 to 8 per million people and are a rare cause of hypertension (0.2%–0.6% of all patients with hypertension),¹³ yet pheochromocytomas make up 4% to 7% of adrenal incidentalomas.¹⁴ Making the diagnosis is key because, when undiagnosed, these tumors are associated with high morbidity and mortality secondary to the uncontrolled catecholamine levels leading to hypertension, heart disease, stroke, and death. Although most tumors are localized, up to 25% can be metastatic and associated with a poor prognosis.¹⁵ There are no good predictors of metastatic disease, and histologic features are not a reliable predictor for aggressive behavior. Large tumor size, extra-adrenal location, *SDHB* germline pathogenic variant, and very high dopamine/methoxytyramine secretion are associated with metastatic disease but are not specific.

Prevalence and Genetic Association With Pheochromocytoma and Paraganglioma

The prevalence of diagnosed pheochromocytoma in patients with hypertension in general medical outpatient clinics is 0.1% to 0.6%¹⁶; the true prevalence may be considerably higher.^{12,16} PPGLs can be

sporadic or familial. Up to 40% of patients with PPGLs will have a germline pathogenic variant in one of over 14 genes known to increase risk of PPGL^{12,15} (Table 40.1). Guidelines recommend that all patients with these tumors be referred for clinical genetic testing because of the high rate of hereditary predisposition¹² and the screening implications for both the patient and family members. The classic tumor suppressor syndromes of neurofibromatosis type 1 (NF1), multiple endocrine neoplasia type 2 (MEN2), and von Hippel–Lindau disease (vHL) were the first discovered to be associated with increased risk of pheochromocytomas, often bilateral, and all are autosomal dominant syndromes (see Table 40.1). The hereditary paraganglioma-pheochromocytoma syndromes are caused by germline pathogenic variants in the *Succinate Dehydrogenase Subunit (SDH)* genes, which make up complex II of the mitochondrial respiratory chain and convert succinate to fumarate in the Krebs cycle. Germline pathogenic variants in any of the four genes *SDHA*, *SDHB*, *SDHC*, or *SDHD* and a cofactor, *SDHAF2*, increase risk for PPGLs. Table 40.1 describes more information. *SDHB* is associated with a higher risk of metastatic disease (~25%) compared with the other *SDHx* genes (<5%).¹⁷ *SDHD* and *SDHAF2* germline pathogenic variants are paternally expressed, meaning a carrier will only develop PPGL if the variant was inherited from their father, with extremely rare exception.¹⁸ Carriers of *SDHx* pathogenic variants are at risk for multiple primary PPGLs, clear cell renal cell carcinoma, gastrointestinal stromal tumors, and pituitary adenomas.¹⁵ *SDHx* carriers must undergo lifelong screening, starting between 6 and 10 years of age with annual biochemistry panels and full-body imaging every 2 to 3 years,¹⁹ usually with MRI rather than CT to minimize radiation exposure.

Clinical Manifestations

Classic symptoms from overproduction of catecholamines include headache, sweating, palpitations, anxiety, and pallor; however, patients may be asymptomatic or present with a variety of symptoms; therefore a high level of suspicion is required to make the diagnosis.²⁰ Although less than 1% of resistant hypertension cases are because of pheochromocytoma/paraganglioma, new or worsening hypertension or diabetes mellitus, with or without symptoms, may be the presenting manifestation. PPGL also may present with symptoms of mass effect because of enlarging tumors. On occasion, a metastatic lesion may be the presenting sign. Physical examination may reveal labile (66%) or persistent (33%) hypertension, and some patients may be normotensive or even hypotensive if epinephrine is the predominant catecholamine secreted. Patients may also present with catecholamine (takutsu) cardiomyopathy. Hypertensive crisis with or without heart failure may be precipitated by surgery, general anesthesia, pregnancy, or rarely on exposure to radiocontrast material. This also may occur after minor or major trauma and from sudden spontaneous release of catecholamines from the tumor or on hemorrhage into it.

Diagnosis

Diagnosis of pheochromocytoma is based on clinical suspicion and biochemical confirmation. Measurement of plasma-free metanephrines or urinary fractionated metanephrines is used for diagnosis or exclusion of pheochromocytoma. Both plasma and urine metanephrine tests have over 90% sensitivity for PPGLs. Plasma metanephrines are favored because of their ease of collection and higher specificity compared with the 24-hour urine tests (ranging from 79%–98% vs. 69%–95%, respectively).¹² Catecholamine and metanephrine measurements are susceptible to false-positive elevations for many reasons. Interfering medications (Box 40.1) are the most common cause. Plasma tests should ideally be performed with the patient resting for 20

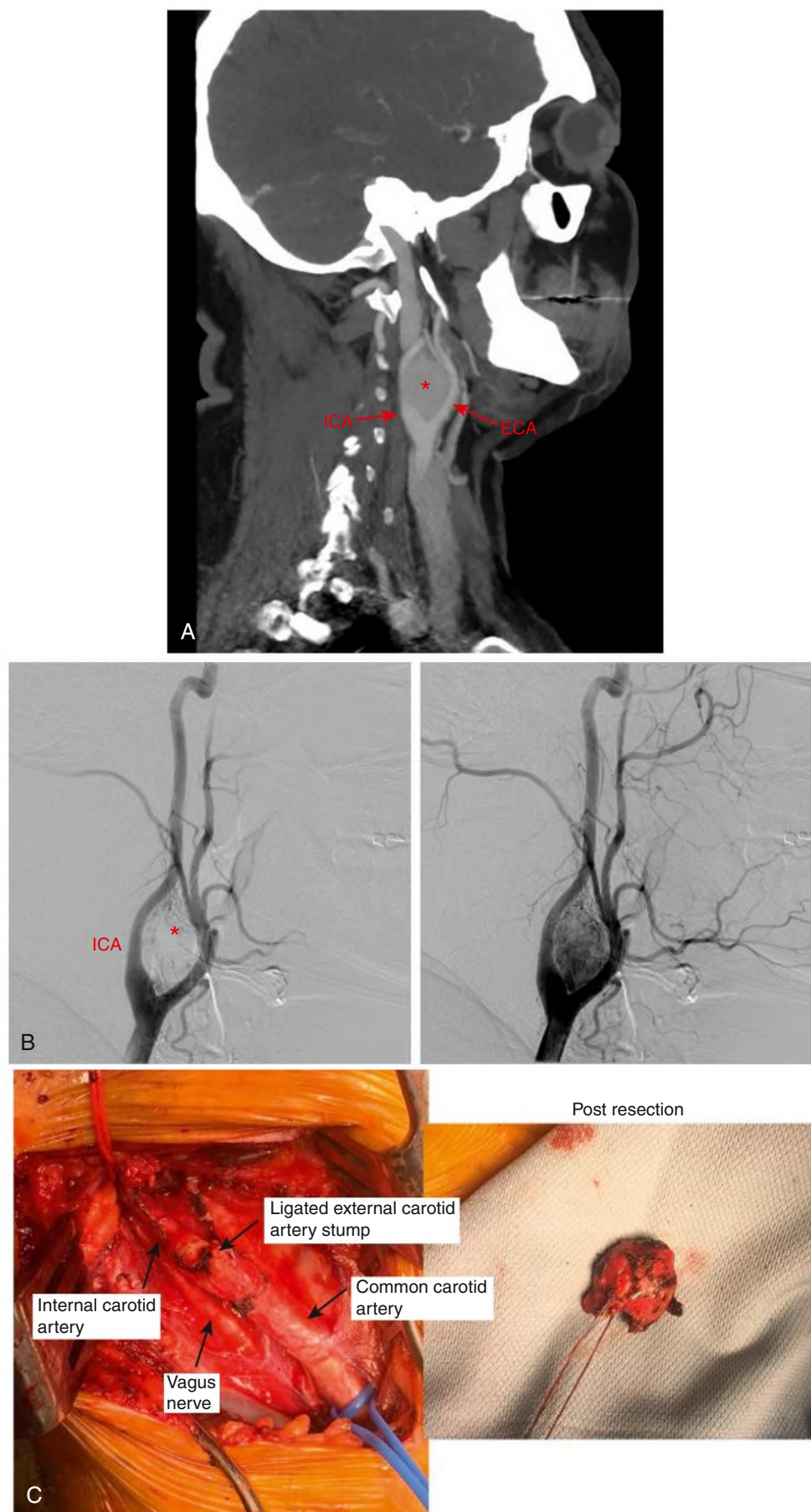


Fig. 40.4 (A) Carotid paraganglioma. (B) Carotid paraganglioma pre- and postembolization before surgery. (C) Carotid paraganglioma intraoperatively and after resection. *ECA*, External carotid artery; *ICA*, internal carotid artery.

TABLE 40.1 Genetic Syndromes and Clinical Features Associated With PPGL

Gene	Syndrome	Primary PPGL location	Mode of inheritance	Clinical Features	Risk of Malignancy	Clinical Pearls
<i>NF1</i>	Neurofibromatosis type 1 (NF1)	Adrenal pheochromocytoma Can be bilateral	Autosomal dominant 1:3000	≥2 of the following clinical criteria: • 6 or more café-au-lait spots • Lisch nodules (benign iris hamartomas) • ≥2 cutaneous neurofibromas • Plexiform neurofibroma • Axillary or inguinal freckling • Optic glioma • Sphenoid dysplasia (thinning of the long bones) • First-degree relative with NF1	Up to 12%	All patients with NF1 and HTN must be screened for pheochromocytoma
<i>RET</i>	Multiple endocrine neoplasia type 2 (MEN2)	Adrenal pheochromocytoma Bilateral pheochromocytoma 50% (synchronous or metachronous)	Autosomal dominant 1:30 000	MEN2A (90%): • Medullary thyroid carcinoma • Pheochromocytoma • Primary hyperparathyroidism MEN2B (10%): • Medullary thyroid carcinoma • Pheochromocytoma • Mucosal neuromas • GI ganglioneuromas • Marfanoid habitus	<5%	Annual biochemical screening for pheochromocytoma • Age 11 in patients with high-risk mutations (including those in codons 634 and 918) • Age 16 for those with moderate risk mutations
<i>VHL</i>	von Hippel–Lindau	Adrenal pheochromocytoma in 10%–20% Bilateral pheochromocytomas	Autosomal dominant 1:36 000	Multiple different tumors and cysts • Pheochromocytoma • Hemangioblastomas of the central nervous system, including retina • Renal cysts • Clear cell renal cell carcinoma • Pancreatic cysts • Pancreatic neuroendocrine tumors • Endolymphatic sac tumors • Epididymal cysts	5%	Annual biochemical screening for pheochromocytoma should begin at age 5 y
Hereditary Paraganglioma-Pheochromocytoma Syndromes						
<i>SDHA</i>		Any location but more commonly in head and neck region	Autosomal dominant	PPGL GIST	May be high at ~12%	Most experts recommend screening starting between age 6 and 10 y with annual biochemistries and full body imaging (MRI preferred) every 3 y
<i>SDHB</i>		Any location but commonly extraadrenal and HNPGL	Autosomal dominant	PPGL Renal cell carcinoma GIST	23%–50%	
<i>SDHC</i>		Any location but commonly HNPGL or thoracic PGL	Autosomal dominant		<5%	
<i>SDHD</i>		Any location but commonly multifocal and often HNPGL	Autosomal dominant and paternal inheritance		<5%	
Other Germline Susceptibility Genes						
<i>TMEM127</i>	Familial PPGL syndrome	Adrenal pheochromocytoma Can be extraadrenal or HNPGL	Autosomal dominant	PPGL Renal cell carcinoma	<5%	Rare cause of PPGL
<i>MAX</i>	Familial PPGL syndrome	Adrenal pheochromocytoma Can be bilateral	Autosomal dominant		Unknown; may be high	Rare cause of PPGL

Continued

TABLE 40.1 Genetic Syndromes and Clinical Features Associated With PPGL—cont'd

Gene	Syndrome	Primary PPGL location	Mode of inheritance	Clinical Features	Risk of Malignancy	Clinical Pearls
<i>HH</i>	HLRCC syndrome	Any location	Autosomal dominant	Leiomyomas of the uterus and skin; renal cell carcinoma	May be high	Rare cause of PPGL
<i>MDH2</i>	Familial PPGL syndrome	Any location	Autosomal dominant			Rare cause of PPGL
<i>EPAS1</i>	Polycythemia paraganglioma syndrome	Any location	Somatic mosaicism	Can have polycythemia and/or somatostatinomas		Rare cause of PPGL

GI, Gastrointestinal; *GIST*, gastrointestinal stromal tumors; *HLRCC*, hereditary leiomyomatosis and renal cell carcinoma; *HNPGL*, head and neck paraganglioma; *HTN*, hypertension; *MRI*, magnetic resonance imaging; *PPGL*, pheochromocytoma and paraganglioma.

BOX 40.1 Medications That Interfere With Screening Tests for Pheochromocytomas and Paragangliomas

Acetaminophen
Levodopa
Monoamine oxidase inhibitors
Selective serotonin reuptake inhibitors
Sympathomimetics
Tricyclic antidepressants
Some β -blockers (especially nonselective)
Some α -blockers (i.e., phenoxybenzamine)
Cyclobenzaprine
Cocaine

minutes in the supine position,¹² although this is not always practical. Most laboratories have different reference ranges for tests drawn in the supine and upright positions to help when interpreting the results. In patients with advanced chronic kidney disease (CKD), there are additional challenges with interpreting test results because catecholamines and their metabolites are renally cleared²¹ and patients with advanced CKD or kidney failure have sympathetic hyperactivity.²² Plasma-free metanephrines appear to be the least dependent on kidney function relative to plasma catecholamines and deconjugated metanephrines but remain higher in patients with kidney disease than hypertensive controls.²¹ Different cutoffs for plasma metanephrine concentration have been suggested based on the severity of CKD, which would reduce false-positive screening results from 7.6% to 5% in patients with CKD stage 3 and from 21.9% to 4.1% in patients with CKD stage 4 or higher.²³ Another study suggested that 24-hour urine normetanephrine measurements can be used to detect PPGLs in people with CKD provided that the glomerular filtration rate (GFR) is greater than 15 mL/min/m².²⁴ End-stage kidney disease patients are often anuric; urinary measurements are not possible, and plasma metanephrines have been found to be approximately twofold higher than hypertensive controls.²¹

Imaging

Once the biochemical diagnosis is secured, the lesion must be localized. Cross-sectional imaging with MRI or CT scan (with contrast) of the abdomen and pelvis is recommended first because most lesions will be within the abdomen and pelvis. For patients with known susceptibility gene mutations, imaging of other anatomic locations may be necessary. Paragangliomas, particularly of the head and neck, that are derived from the parasympathetic chain are often nonsecretory;

therefore, if suspected, imaging should be performed regardless of biochemical testing results. If patients with a known head and neck paraganglioma have elevated metanephrines, abdominal/pelvic imaging studies must be performed to evaluate for additional sympathetic-derived primary PPGLs.

Imaging with ¹²³I-metaiodobenzylguanidine (MIBG) should be reserved for patients in whom cross-sectional imaging did not reveal a tumor despite markedly abnormal biochemical testing or patients with metastatic disease to assess if the lesions are amenable to treatment with ¹³¹I-MIBG.¹² ⁶⁸GA DOTATATE PET/CT or ¹⁸F-FDG PET/CT scanning is preferred (particularly in patients with *SDHB* germline pathogenic variants) over ¹²³I-MIBG imaging for diagnosis of metastatic disease because of higher sensitivity and specificity.^{25–28}

Treatment

Once a PPGL has been localized, the patient should be prepared for surgery with a collaborative team approach by the surgeon, anesthesiologist, and physician. Perioperative blockade is important to reduce the morbidity and mortality associated with tumor resection (down to 0%–2%²⁹) and should be used before other surgical procedures and biopsies and considered before treatment for metastatic disease such as chemotherapy, radiation, and ¹³¹I-MIBG therapy, particularly when catecholamine levels are very elevated. Independent of medication regimen, treatment should also include a high-sodium diet and increased fluid intake. The typical drugs and dosing regimens are shown in Table 40.2.

Medications

α -Blockers

Secretory PPGLs cause α -receptor activation in response to excess catecholamine secretion leading to severe vasoconstriction, which can result in hypertension, arrhythmias, and myocardial ischemia. Therefore, selective or nonselective α -blockers are recommended as first-line therapy in perioperative management.¹² A prospective randomized controlled trial suggested no major differences in using phenoxybenzamine (nonselective) versus doxazosin (α -1 selective) α blockade.³⁰ The nonselective α -blocker phenoxybenzamine is a noncompetitive inhibitor that covalently (irreversibly) binds to α -1 and α -2 receptors and provides more complete blockade of α receptors. This significantly lowers the risk of an intraoperative hypertensive crisis but can also cause hypotension once the tumor is resected, so vasopressors and IV fluids may be required for 24 to 48 hours postoperatively.

Selective α -1 receptor blockers include doxazosin, terazosin, and prazosin. These competitive inhibitors are relatively short-acting, and their effects can be overcome by the excess catecholamine released during surgery, potentially causing an intraoperative hypertensive crisis. However, their shorter half-life results in less postoperative hypotension.

TABLE 40.2 Common Medications for Perioperative Blockade of Patients With Pheochromocytoma and Paraganglioma

Drug	Mechanism of Action	Common Dosing	Common Side Effects
Nonselective α-Blocker			
Phenoxybenzamine	Noncompetitive antagonist	10 mg 2–3 times daily (maximum 60–100 mg/d)	Orthostasis, nasal congestion
Selective α-Blockers			
Doxazosin	Competitive antagonist	4–8 mg twice daily	Orthostasis, dizziness
Prazosin	Competitive antagonist	5 mg three times daily	Orthostasis, dizziness
Terazosin	Competitive antagonist	1–4 mg once daily	Orthostasis, dizziness
Calcium Channel Blockers			
Nicardipine	Dihydropyridine long acting	30 mg twice daily	Headache, edema, vasodilation
Amlodipine	Dihydropyridine long acting	5–10 mg daily	Headache, edema, palpitations
Tyrosine Hydroxylase Inhibitor			
Metyrosine	Decreases catecholamine production	250–500 mg 4 times daily (dose escalated every 2 days)	Severe lethargy, extrapyramidal neurologic side effects and gastrointestinal upset
β-Blockers			
Metoprolol tartrate (selective β -1 blocker)	Used to treat reflex tachycardia only after full α -blockade achieved	25–50 mg 1–2 times daily	Fatigue, dizziness
Atenolol (nonselective β -1 blocker)	Used to treat reflex tachycardia only after full α -blockade achieved	25–100 mg/d	Fatigue, dizziness

β -Blockers

Preoperative coadministration of β -adrenergic receptor blockers is indicated to control reflex tachycardia only after administration of α -adrenergic receptor blockers and sufficient volume expansion to avoid the potential for hypertensive crisis because of unopposed stimulation of α -adrenergic receptors. This tachyarrhythmia is a desired side effect indicating complete α -blockade has been achieved.

Calcium Channel Blockers

Calcium channel blockers are second-line therapy or add-on therapy.¹² No prospective studies compare α -blockade and calcium channel blockers for preoperative blockade.

Metyrosine

Methyl-paratyrosine (metyrosine) inhibits catecholamine synthesis and may be used in combination with α -adrenergic receptor blockers for a short period before surgery to further stabilize BP or to help control symptoms related to catecholamine excess in those with metastatic PPGL. Metyrosine is infrequently used because it is expensive, its efficacy is controversial, it is not readily available, and it has significant side effects.

Acute Hypertensive Crisis

Acute hypertensive crisis can be the presenting sign in patients with an undiagnosed PPGL and can occur in patients with a known tumor. Intravenous (IV) α -blockade with phentolamine can be used to control BP, with or without other IV vasodilators such as sodium nitroprusside or nicardipine.

Surgery

Surgical resection is the treatment of choice for PPGLs. Laparoscopic surgery for PPGL is associated with better outcomes than open adrenalectomy³¹ for small PPGLs. Cortical-sparing surgery should be attempted in patients with or without genetic predisposition for bilateral adrenal pheochromocytomas (MEN2 and vHL) because it

may avoid the need for lifelong glucocorticoid and mineralocorticoid replacement. However, there is a risk of recurrence with cortical-sparing surgery. Continuous hemodynamic and cardiovascular monitoring is required during surgery and perioperatively. Postoperatively, patients may require BP support with fluids, colloids, and sometimes α -adrenergic agonists for 24 to 48 hours. BP usually returns to normal within a few days of surgery, but patients may remain hypertensive, particularly if they have chronic underlying hypertension or widespread metastatic disease.

Follow-up

All patients should have plasma metanephrines checked about 4 to 6 weeks after surgery; if levels remain elevated, this may indicate residual or metastatic disease. All patients should have annual plasma metanephrines levels checked for life because recurrence or metastatic disease can occur even 20 years later, and patients also can develop additional primary tumors. All patients with PPGL should be referred for genetic testing because of the high rate of inherited disease (up to 40%).¹² Follow-up imaging is not required for most patients with complete sympathetic PPGL resection unless they have a known germline susceptibility gene mutation. Imaging also should be considered if catecholamines remain elevated or if metastatic disease is suspected.

Pheochromocytoma and Paraganglioma in Pregnancy

PPGL in pregnancy can be dangerous, even fatal, for both the mother and fetus. Perioperative blockade is the same as for nonpregnant patients; however, timing of surgery is often tricky. It is usually recommended to proceed with surgical resection around 18 to 22 weeks of pregnancy. If diagnosis is not made until the third trimester, it is recommended to perform a cesarean section and surgical resection of the PPGL at the same surgery. Spontaneous labor and delivery should be avoided. In patients with genetic predisposition, screening with plasma metanephrine levels is recommended when contemplating conception and/or when pregnancy is confirmed to avoid morbidity from an undetected PPGL. If patients develop a hypertensive crisis during pregnancy, treatment is

the same as for nonpregnant patients except nitroprusside should be avoided because of the risk of cyanide toxicity in the fetus. A recent large multicenter, retrospective study of patients with PPGL and pregnancy included 249 pregnancies in 232 patients with PPGL, and 66% had an underlying germline mutation.³² Adverse outcomes were associated with unrecognized PPGL during pregnancy, abdominal or pelvic tumor location, and very high catecholamine levels at least 10 times the upper limit of the normal range. For patients diagnosed during pregnancy, α -adrenergic blockade therapy was associated with fewer adverse outcomes, and the outcome was similar if surgery was performed during pregnancy or delayed until after pregnancy.

Metastatic Pheochromocytoma and Paraganglioma

PPGLs should be described as metastatic or nonmetastatic PPGL rather than malignant or benign.¹³ Common metastatic sites include lymph nodes, liver, bones including the skull, lung, and peritoneum. Metastases occur in approximately 10% of pheochromocytomas and 20% of paraganglioma and are more likely to occur in large tumors, extra-adrenal paragangliomas, and patients with *SDHB* mutations.³³ Metastases can be detected at the time of primary tumor diagnosis or even 20 years later and are associated with 5-year survival of 50%.³³ Unfortunately, there are no reliable histopathologic markers to predict metastatic disease.

Treatment Options

Treatments for metastatic disease can slow disease progression, but none are curative. Surgical debulking is still the best option when tumor burden is low. If the tumor is MIBG avid, ¹³¹I-MIBG therapy can be used. Recently high-specific-activity ¹³¹I-MIBG was US Food and Drug Administration (FDA) approved for treatment of patients with metastatic PPGL. Peptide receptor radionuclide therapy (PRRT) using radiolabeled somatostatin analogs is FDA approved for gastrointestinal and pancreatic neuroendocrine tumors and is in clinical trials for PPGL. External beam radiation therapy and proton therapy often are used for bone metastases and control of unresectable head and neck paragangliomas. Cytotoxic chemotherapy with cyclophosphamide, vincristine, and dacarbazine (CVD) can control disease but is not curative.

ADRENAL INCIDENTALOMA

Definition and Epidemiology

Adrenal incidentaloma refers to the incidental discovery of an adrenal mass in the absence of any prior suspicion of adrenal disease.^{34,35} The prevalence of unsuspected adrenal masses of adults undergoing high-resolution CT or MRI abdominal studies is approximately 1% in those younger than 30 years, rising to 10% in the elderly.³⁴

Hypertension is more common in those with incidentalomas (40%) than in the general population. About 75% prove to be nonsecretory benign adenomas. However, an important minority show hormone secretion, including aldosterone-producing adenomas (2.5%, frequently normokalemic), pheochromocytoma (7%, often normotensive), and, most often, cortisol-secreting tumors (12%).³⁴

Subclinical Cushing syndrome (SCS), now referred to as mild autonomous cortisol secretion (MACS) without clinical manifestations of Cushing syndrome, is the most frequent hormonal abnormality detected in patients with adrenal incidentalomas. Cortisol excess should be ruled out by performing a 1-mg overnight dexamethasone suppression test (DST). If the 8 AM serum cortisol concentration after 1 mg of dexamethasone given at midnight the evening before is greater than 1.8 μ g/dL, this suggests hypercortisolism. Typically, a level greater than 5 μ g/dL is considered overt hypercortisolism, whereas levels

between 1.8 and 5 μ g/dL are considered MACS. In addition, DHEAS is low in adrenal cortisol excess. A baseline morning ACTH and cortisol can confirm if this hypercortisolism is ACTH independent (ACTH should be low) and from an adrenal nodule. Hypercortisolism, even mild hypercortisolism, is frequently complicated by hypertension (~65% of cases), diabetes (~33%), obesity, and osteoporosis and carries increased risk for CV events and related mortality.^{36,37} A recent retrospective study of 632 patients with either MACS ($n = 212$) or nonsecreting adenomas ($n = 420$) demonstrated a higher prevalence of atrial fibrillation in those with MACS (8.5%) compared with the nonsecreting group (3.1%).³⁸

Adrenal cortical carcinoma (ACC) is rare, affecting 1 to 2 per million people³⁹ and accounts for a very small percentage of adrenal tumors, but is deadly with 5-year survival of less than 30%. Most ACC are greater than 4 cm on imaging and may or may not be functional (e.g., with evidence of Cushing syndrome, primary aldosteronism, and/or virilization).

Management

Investigation of adrenal incidentalomas should address two key issues: whether the tumor is malignant and whether it is hormonally active. Every adrenal incidentaloma should be evaluated for pheochromocytoma (plasma-free metanephrines or 24-hour urine fractionated metanephrines), glucocorticoid excess (early-morning plasma cortisol concentration after 1 mg dexamethasone given the night before), and, if the patient has hypertension, also evaluated for primary aldosteronism (plasma renin and serum aldosterone levels), as outlined here and in [Chapter 39](#). Tumors with an appearance that is worrisome for malignancy (e.g., noncontrast Hounsfield units [HU] >10), tumors that are growing, tumors more than 4 to 6 cm in diameter, and those that are functional should be surgically resected.^{34,35} Adrenal incidentaloma guidelines suggest benign-appearing (HU <10), small adenomas (less than 4–6 cm) that are nonfunctional no longer need imaging or biochemical follow-up,³⁴ although some experts suggest even nonfunctional nodules may have long-term cardiometabolic consequences and should have some subsequent follow-up.

RENIN-SECRETING TUMOR

Definition

Primary renin-secreting tumors are rare, and less than 200 cases have been reported.⁴⁰ Diagnostic criteria include an elevated plasma renin or prorenin level, which decreases on removal of the tumor, and demonstration of renin within the tumor. Most renin-secreting tumors are caused by benign renal juxtaglomerular cell tumors (JGCT) ranging from 5 mm to 6 cm in diameter (typically 2–4 cm). These tumors occasionally occur with nephroblastomas, renal cell carcinomas, and extrarenal neoplasms (bronchial or pancreatic carcinoma, ovarian tumors, carcinoma of ileum or colon, soft tissue sarcomas, orbital hemangiopericytoma).

Etiology and Pathogenesis

Autonomous hypersecretion of renin results in high circulating levels of Ang II that increase arterial pressure. Secondary hyperaldosteronism and hypokalemia result from stimulation of the adrenal glomerulosa by Ang II. High Ang II levels also induce hyponatremia in a minority of patients, by stimulation of thirst and arginine vasopressin (AVP) secretion, together with a direct renal water-retaining action of the peptide.⁴⁰

Clinical Manifestations

Cases show female predominance; 75% of patients are aged less than 30 years, presenting usually with severe, occasionally paroxysmal

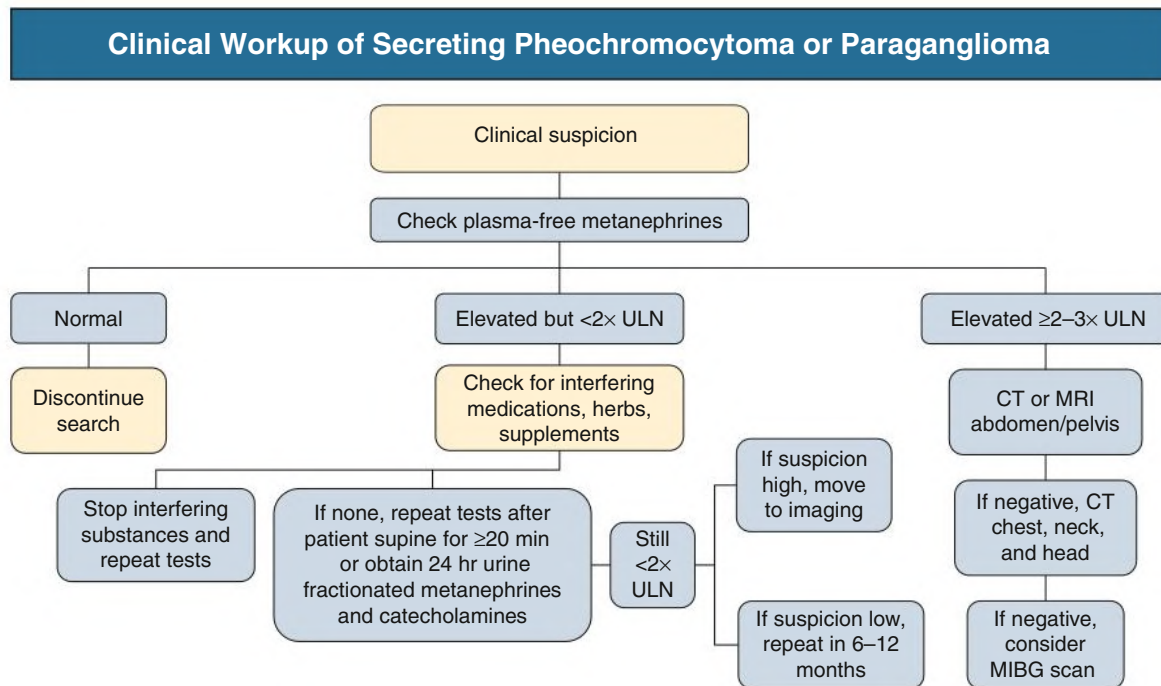


Fig. 40.5 Clinical workup of secreting pheochromocytoma or paraganglioma. *CT*, Computed tomography; *MIBG*, metaiodobenzylguanidine; *MRI*, magnetic resonance imaging; *ULN*, upper limit of normal.

hypertension (average 206/131 mm Hg), hypokalemia (<3.0 mmol/L in ~70% of cases), proteinuria (>0.4 g/day in ~50% of patients), and, in a minority, hyponatremia.⁴¹ GFR is normal or high. BP may decrease substantially with the first dose of angiotensin-converting enzyme (ACE) inhibitor or angiotensin receptor blocker (ARB).

Pathology

Renin-secreting tumors are encapsulated and tan or grayish yellow, with scattered hemorrhages. These masses consist largely of polygonal or spindle cells in close contact with capillary and sinusoidal vessels and contain cytoplasmic renin granules.⁴⁰

Diagnosis and Differential Diagnosis

Patients presenting with hypertension and hypokalemia together with elevated renin (and prorenin) and aldosterone levels may harbor a renin-secreting tumor, most often a JGCT.

CT or MRI angiography or renal arteriography typically identify JGCTs as radiolucent, expansile, homogeneous, solid renal cortical neoplasms, usually 2 to 4 cm in diameter (Fig. 40.5).⁴¹ CT and MRI scans showing an isodense or hypodense lesion with little or no enhancement after injection of contrast material have proved helpful in the provisional localization of these tumors. Bilateral, simultaneous renal vein sampling may enable lateralization of the tumor. Because renal blood flow to the culprit kidney is not impaired, however, a renin ratio of more than 1.2:1 between the two renal veins may not be present, in contrast to unilateral renal artery stenosis, in which reduced blood flow to the stenosed kidney and renin oversecretion often lead to an elevated renal vein renin ratio. Selective segmental renal vein renin sampling may help localize the tumor. When no renal lesion can be visualized and lateralization of renin secretion is not evident, an extrarenal renin-secreting lesion must be considered and sought by appropriate radiographic investigations and venous sampling for renin measurements.

Apart from renal artery stenosis or occlusion, it may be necessary to exclude other renin-producing lesions, including Wilms tumor, renal carcinoma, neuroblastoma, hepatocellular carcinoma, and

pheochromocytoma, which can either secrete renin or stimulate renal renin production.

Treatment

Preoperative BP control is achieved using an ACE inhibitor or ARB, introduced cautiously to avoid first-dose hypotension. For JGCTs, local excision is advisable, if possible, to preserve nephrons. When doubt exists, an intraoperative frozen section will differentiate a benign JGCT from malignant lesions and guide surgery. Removal of a JGCT results in the return of renin and aldosterone levels to normal. BP decreases rapidly but not always to normal if there is a history of primary hypertension.⁴²

ACROMEGALY

Definition and Epidemiology

Acromegaly is caused by excessive circulating growth hormone (GH) originating from a somatotroph (or somatomammotroph) adenoma of the pituitary in more than 95% of cases.⁴³ Rare causes include hypothalamic or neuroendocrine tumors secreting GH-releasing factor (GHRH) and, even more rarely, hemopoietic, breast, bronchial, and abdominal tumors may secrete GH. Multiple endocrine neoplasia type 1 is associated with pituitary adenomas, with somatotroph adenomas being the second-most common of the secreting tumors. Acromegaly may appear in other familial settings (e.g., McCune-Albright syndrome, Carney complex) or as an isolated disorder in familial isolated pituitary adenoma.

Acromegaly is rare, with an incidence of 6 to 8 cases per million population per year.⁴⁴ The median age of diagnosis is 47 years. Hypertension is more common (~40%) in persons with acromegaly than in the general population, especially in female and older patients.⁴⁵ Patients with acromegaly who have additional hypopituitarism or advanced cardiomyopathy may have BP reduction masking before hypertension. Most somatotroph adenomas are macroadenomas resulting in compression of the normal pituitary and hypopituitarism in 75% of cases.



Fig. 40.6 Facial features of acromegaly with enlargement of brow, nose, and jaw.

The pathogenesis of hypertension in acromegaly is complex but reflects sodium retention and volume expansion associated with an inappropriate compensatory response. Total exchangeable sodium, total body water, and extracellular fluid volume are increased. Volume expansion should suppress plasma renin levels, but although levels are low, they are not consistent with the sodium status. Aldosterone levels are also normal or only slightly suppressed. The kidneys are enlarged and GFR is increased, but sodium balance is not corrected unless the acromegaly is cured.

Clinical Manifestations

Excess circulating GH stimulates production of insulin-like growth factor 1 (IGF-1) from the liver. IGF-1 mediates the biologic effects of GH. Acromegaly is characterized by enlargement of the jaw (macrogathia), skull (Fig. 40.6), hands (Fig. 40.7), and feet because of bone and soft tissue overgrowth. Other symptoms result from local effects of an expanding pituitary tumor and include visual field defects and headache. The manifestations of disease are gradual, slowly progressing over many years, and diagnosis may be delayed until dysmorphic features are well developed, typically in the fourth to fifth decade. Other signs and symptoms include headache (40%), excessive sweating (50%), loss of libido (35%), amenorrhea (45%), carpal tunnel syndrome (25%), diabetes mellitus (19%), and visual field defects (5%).⁴³ Thyroid enlargement occurs in 50% of patients and thyrotoxicosis in 6%; hirsutism occurs in 24% of women and galactorrhea in 10%. Other complications of acromegaly include cardiac hypertrophy, systolic and diastolic dysfunction, heart failure, arrhythmia, sleep apnea, osteoarthritis and osteopenia, and disturbed calcium metabolism.⁴⁶ Acromegaly is associated with increased risk of malignancy, including a twofold increase in colorectal cancer.

Diagnosis

Clinical suspicion should be raised by symptoms and signs. Box 40.2 lists appropriate tests for acromegaly. Elevated levels of IGF-1 and GH, especially with lack of suppression of GH on oral glucose tolerance testing, are strongly suggestive of the diagnosis. MRI of the pituitary and visual field testing for those with macroadenomas are necessary to define the tumor extension.

Treatment

Treatment aims to lower IGF1 (GH) levels so that symptoms and signs resolve and metabolic abnormalities improve. However, bony



Fig. 40.7 Radiograph of the Hand in Acromegaly. “Arrowhead” distal phalanges, expanded joint spaces, and increased soft tissue can be seen.

BOX 40.2 Diagnostic Tests for Acromegaly

- Screen with serum insulin-like growth factor-1. If elevated:
 - Serum growth hormone responses to glucose tolerance test to confirm
 - Assessment of full pituitary function (thyroid-stimulating hormone, T4 [thyroxine], prolactin, adrenocorticotrophic hormone, cortisol, luteinizing hormone, follicle-stimulating hormone, estradiol/testosterone)
 - Magnetic resonance imaging of pituitary fossa; if macroadenoma, visual field testing

overgrowth generally does not reverse. Resection of the pituitary adenoma by transsphenoidal surgery is the treatment of choice. Surgical cure (defined by normalization of plasma IGF-1 and a nadir oral glucose tolerance test GH of $< 2 \mu\text{g/L}$) is achieved in 81% to 100% of microadenomas and 45% to 68% of macroadenomas.⁴³ Irradiation and drug therapy are valuable when complete removal of tumor tissue is not possible (approximately one-third of acromegalic cases overall) and when surgery is contraindicated. Somatostatin analogs are first-line medical therapy because they can decrease the size of the residual tumor in addition to lowering GH secretion. If IGF1 levels have not normalized, pegvisomant, a GH receptor antagonist, can be added or may be used as first-line therapy.^{43,47} Pegvisomant does not shrink the tumor. Dopaminergic agents, such as bromocriptine and cabergoline, may be helpful to reduce GH levels in mild cases but are not strongly effective. Radiotherapy is used in cases with persistent active disease despite all other treatments but may not exert its full effect for months or years. Patients must be monitored regularly for pituitary function because hypopituitarism may occur late after treatment (even 10 years out) and necessitate hormone replacement in about 40% of patients.⁴⁸

Management of Hypertension in Acromegaly

Surgical removal of the pituitary adenoma with normalization of GH levels may reduce BP, but most patients with acromegaly will continue to require antihypertensive therapy. Antihypertensive treatment

requires a diuretic, given the volume-expanded state. Additional anti-hypertensive agents are frequently required, and both calcium channel blockers and ACE inhibitors may be effective. β -Blockers also may be used, although theoretically such agents may increase GH concentration.

HYPOTHYROIDISM

Definition and Epidemiology

Hypertension is more common in hypothyroid patients than in the general population. The pathogenesis of the hypertension is multifactorial and associated with both increased total body sodium and increased peripheral vascular resistance.

Hypothyroidism is associated with increased aortic stiffness, loss of sensitivity to vasoconstrictors, and impaired endothelial function with loss of endothelial-dependent vasodilation plus reduced vasodilatory responses to NO donors.⁴⁹ Hypothyroidism is associated with increases in arterial pressure, plasma catecholamines, aldosterone, and cortisol.⁵⁰⁻⁵² The relationship between plasma catecholamine levels and BP is enhanced in hypothyroidism.⁵³ Hypertension develops despite a low cardiac output.

Thyroid replacement therapy corrects the electrolyte, hemodynamic, and hormone changes and cures the hypertension in most patients.

Clinical Features

Bradycardia, mild hypertension with narrowed pulse pressure, and muffled heart sounds are the most frequent signs in overt hypothyroidism. Other features include weakness, dry skin, lethargy, slow slurred speech, cold intolerance, thick tongue, facial puffiness, coarse and thinning hair, failing memory, constipation, and weight gain with reduced appetite. In extreme cases, myxedema and coma ensue. Coronary heart disease is common, with dyslipidemia and hypertension accelerating atherogenesis.

Diagnosis

Hypothyroidism should be considered in any patient with hypertension. Because the clinical manifestations of hypothyroidism are often difficult to elicit, especially in elderly patients, thyroid function tests, including thyroid stimulating hormone (TSH), should be performed. In patients with primary hypothyroidism who remain hypertensive despite full thyroxine replacement therapy, primary hypertension is likely also present. For those patients with known heart failure or angina who have overt or long-standing hypothyroidism, replacement thyroxine therapy should be given initially with low doses and increased slowly over time to minimize the chance of precipitating myocardial ischemia.

HYPERTHYROIDISM

Definition and Epidemiology

Hypertension is common in hyperthyroidism, with a prevalence of 60% in toxic adenoma and approximately 30% in Graves disease.²

Clinical Features

The clinical features depend on the underlying cause of the hyperthyroidism, the severity of the disorder, rapidity of onset, age of the patient, and concomitant disease. Abnormalities may be evident in the CV system (tachyarrhythmias, heart failure), skin (increased sweating, increasing pigmentation with vitiligo), eyes (lid lag, exophthalmos),

nervous system (hypertension, nervousness), gastrointestinal system (increased appetite yet weight loss, diarrhea), and muscles (proximal weakness).

Hypertension in hyperthyroidism is associated with an increased pulse pressure and elevated systolic BP with normal or low diastolic BP.⁵⁴ It may be observed in both postpartum thyrotoxicosis and neonatal thyrotoxicosis. Elevation of diastolic BP is unusual unless there is concomitant primary hypertension.

The hemodynamic characteristics in hypertension of thyrotoxicosis are an increased cardiac output (by 50%–300%), increased myocardial contractility, tachycardia, decreased peripheral vascular resistance, and expanded blood volume. These indices return to normal in most patients on achieving the euthyroid state. Interestingly, catecholamine levels tend to be low (inversely to hypothyroid hypertension), and there is no sympathetic overactivity. The RAS tends to be activated and the aldosterone levels increased in hyperthyroidism, which may contribute to the development of systolic hypertension. Suspicion of hyperthyroidism should be high in the elderly patient with hypertension and a high pulse pressure, particularly if there is also atrial fibrillation. Such patients are prone to developing cardiac failure, in which case the increased systolic arterial pressure will diminish, masking previous hypertension. Hypertension with a high pulse pressure, although typical of hyperthyroidism, is observed in many elderly patients with primary hypertension because of the loss of compliance of the aorta with aging.

Diagnosis and Treatment

The diagnosis is confirmed by thyroid function tests, including measurement of TSH, and when suppressed, adding a T4 and T3. β -Blockers are effective first-line therapy for hyperthyroidism-associated hypertension because they prevent hyperthyroidism-associated arrhythmias, such as tachycardia and atrial fibrillation. Treatment of hyperthyroidism, whether by antithyroid drugs, surgery, or radioiodine, often will promptly normalize the increased systolic arterial pressure, unless concomitant primary hypertension is present.

PRIMARY HYPERPARATHYROIDISM

The incidence of primary hyperparathyroidism (PHPT) is approximately 20 per 100,000 person-years.^{2,55} Full-blown PHPT may feature severe hypercalcemia, nephrolithiasis, acute and chronic kidney disease, and fractures, but patients are usually asymptomatic. The prevalence of hypertension in PHPT approaches 50% with an independent association between plasma ionized calcium levels and the presence of hypertension. In those with MEN2, the hypertension could be a sign of coincident pheochromocytoma.

Indications for surgery in PHPT include serum calcium at least 1 mg/dL above the upper limit of normal, presence of renal stones, decreased renal function with creatinine clearance less than 60 mL/min, elevated urinary calcium greater than 400 mg/day, osteoporosis or vertebral fracture, or age less than 50 years.⁵⁶ Generally, mild PHPT in those aged older than 50 years is not an indication for parathyroidectomy, and management involves serial follow-up with monitoring of plasma calcium, renal function, and bone density. Coincident hypertension is not an accepted indication for parathyroidectomy and is managed according to standard guidelines with the notable difference that thiazide diuretics (which may increase plasma calcium) should be avoided. There is some evidence that surgical cure of PHPT and restoration of eucalcemia restores vascular function and BP toward normal.⁵⁵

SELF-ASSESSMENT QUESTIONS

1. Each form of endocrine hypertension:
 - A. always manifests typical signs and symptoms.
 - B. can always be cured if correct specific therapies are applied.
 - C. is usually familial.
 - D. may present as hypertension resistant to multiple drugs.
 2. Initial screening tests for Cushing syndrome include which of the following?
 - A. CT or MRI of pituitary and adrenal glands
 - B. 1 mg dexamethasone suppression test
 - C. High-dose dexamethasone suppression test
 - D. Bilateral inferior petrosal sinus sampling for ACTH
 3. Pheochromocytomas/paragangliomas are hereditary in what percent of cases?
 - A. <5%
 - B. 10% to 15%
 - C. 30% to 40%
 - D. >70%
-

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